

ICT OPERATOR LEVEL 4

IT/OS/ICTA/CR/01-04/4

ICT Operations and Support

Series:



**TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION
COUNCIL (TVET CDACC)**

PRACTICAL ASSESSMENT

INSTRUCTIONS TO THE ASSESSOR:

1. This assessment is to take place in the prescribed order as arranged in the tool.
2. Capture clear **photographs** and/or **videos** of each candidate's work as they perform the tasks and Label all media files with: Candidate Registration Number, Unit Code, Practical Session Number, and Date.
3. Record candidate scores and assessor remarks in the observation checklists for each session.
4. Store all completed checklists, media files, and candidate work files in a secure digital/physical folder per candidate.

1. CANDIDATE & ASSESSOR DETAILS

Candidate Name:		Reg. Code:	
Assessor Name:		Date:	
Venue:			

2. PROJECT BRIEF

In this project, you will be required to demonstrate competence in all four core units by completing three hands-on sessions (each ~4 hours) and an oral assessment (~1 hour). You will work with computer hardware, software applications, networking equipment, and troubleshooting scenarios to show comprehensive ICT operator skills.

3. EQUIPMENT LIST (PER CANDIDATE)

Item	Qty	Provided By
Desktop computer (complete system)	1	Centre
Additional RAM module	1	Centre
Network cables (Cat 5e/6)	3	Centre
RJ45 connectors and crimping tools	1 set	Centre
Network switch/hub	1	Centre
Router (basic)	1	Centre
Cable tester	1	Centre
Screwdriver set (Phillips & flat)	1 set	Centre
Anti-static wrist strap	1	Centre
Multimeter	1	Centre
External storage device (USB)	1	Centre
Printer	1	Centre
Microsoft Office suite or equivalent	Licensed	Centre
Database software (MS Access/MySQL)	Licensed	Centre
Graphic design software	Licensed	Centre
Sample documents for processing	1 set	Centre
PPE (gloves, dust coat)	1 set	Candidate

4. SECTION 1 (SESSION 1 -- 4 HOURS): COMPUTER ESSENTIALS & OPERATIONS

Elements Covered

1. Manage computer devices
2. Manage desktop settings
3. Perform file management
4. Manage computer software
5. Process computerized word documents
6. Manipulate computerized spreadsheets

4.1 Tasks

Computer Hardware Management:

- Safely disassemble computer case and identify all major components
- Remove and reinstall RAM module following proper procedures
- Reassemble computer and perform successful boot process
- Connect and configure peripheral devices (printer, external storage)

Desktop and File Management:

- Customize desktop settings (icons, date/time, display properties)
- Create organized folder structure with proper naming conventions
- Perform file operations (copy, move, backup to external media)
- Set up file protection and access permissions

Document Processing:

- Create a comprehensive word document with tables, images, and formatting
- Perform mail merge operation with provided data
- Generate table of contents and list of figures
- Create and format a spreadsheet with formulas and functions
- Generate charts from spreadsheet data

4.2 Observation Checklist

No.	Performance Criterion	Max Marks	Awarded
1	Computer hardware devices selected and identified correctly	5	
2	Safe disassembly and reassembly procedures followed	5	
3	RAM module removed and installed properly	5	
4	Computer boots successfully after reassembly	5	
5	Peripheral devices connected and configured correctly	5	
6	Desktop icons and settings customized appropriately	4	
7	File and folder structure created with proper organization	4	
8	File operations performed successfully across different media	4	
9	File protection and permissions applied correctly	4	
10	Word document created with all required elements	5	
11	Mail merge operation completed successfully	4	
12	Table of contents and list of figures generated correctly	4	
13	Spreadsheet created with appropriate formulas and functions	5	
14	Charts generated accurately from spreadsheet data	5	
	Total Section 1	70	

5. SECTION 2 (SESSION 2 -- 4 HOURS): DATABASE, PRESENTATION & NETWORKING

Elements Covered

1. Maintain computerized databases
2. Prepare PowerPoint presentations
3. Manipulate graphic applications
4. Terminate computer network cables
5. Connect computer network cables

5.1 Tasks

Database Operations:

- Design and create a database schema with multiple related tables
- Input data and establish relationships between tables
- Create queries to retrieve and manipulate data
- Generate a report from database information

Presentation and Graphics:

- Create a multi-slide PowerPoint presentation with animations and transitions
- Include charts, images, and multimedia elements
- Design a simple graphic using available software
- Print presentation materials in different formats

Network Setup:

- Terminate network cables using proper color coding and crimping techniques
- Test cables using cable tester to ensure connectivity
- Set up basic network using switch and router
- Configure network settings and establish connections between computers

5.2 Observation Checklist

No.	Performance Criterion	Max Marks	Awarded
1	Database schema designed with proper table relationships	6	
2	Data input and manipulation performed accurately	5	
3	Database queries created to retrieve specific information	5	
4	Database report generated with appropriate formatting	4	
5	PowerPoint presentation created with all required elements	5	
6	Animations and transitions applied effectively	4	
7	Multimedia elements integrated properly	4	
8	Graphic design created meeting basic requirements	4	
9	Network cables terminated with correct color coding	6	
10	Cable testing performed and results documented	5	
11	Network devices configured according to standards	6	
12	Network connection established successfully	6	
	Total Section 2	60	

6. SECTION 3 (SESSION 3 -- 4 HOURS): MAINTENANCE & TROUBLESHOOTING

Elements Covered

1. Perform computer troubleshooting
2. Repair faulty components
3. Test computer component functionality
4. Perform computer maintenance
5. Perform computer network maintenance
6. Perform online jobs and collaboration

6.1 Tasks

Computer Repair and Maintenance:

- Diagnose provided computer problems using systematic troubleshooting
- Identify and replace/repair faulty components following safety procedures
- Perform POST test and component functionality testing
- Create maintenance schedule and perform routine maintenance tasks
- Generate comprehensive maintenance and repair reports

Network Maintenance:

- Monitor network performance and identify potential issues
- Perform network troubleshooting using appropriate tools
- Optimize network settings for improved performance
- Document network maintenance procedures

Online Operations:

- Set up online collaboration tools and demonstrate usage
- Create professional online profiles for job platforms
- Demonstrate online task execution and file sharing
- Show proper digital identity management practices

6.2 Observation Checklist

No.	Performance Criterion	Max Marks	Awarded
1	Computer problems identified through systematic assessment	6	
2	Appropriate solutions determined for identified problems	5	
3	Safety procedures observed during repair operations	5	
4	Faulty components repaired/replaced following manufacturer guidelines	6	
5	POST test and component functionality testing performed	5	
6	Computer maintenance scheduled and executed properly	5	
7	Maintenance reports generated with accurate documentation	4	
8	Network monitoring and troubleshooting performed effectively	5	
9	Network optimization techniques applied correctly	4	
10	Online collaboration tools set up and demonstrated	3	
11	Professional online profiles created appropriately	3	
12	Digital identity management practices demonstrated	4	
	Total Section 3	55	

7. SECTION 4: ORAL ASSESSMENT (25 MARKS)

Assessor to award marks for each correct response by the candidate in the table below:

Q#	Question	Expected Key Points	Max Marks	Awarded
1	Explain the importance of anti-static precautions when handling computer components.	Prevent ESD damage, use wrist straps and mats, proper grounding	3	
2	What are the key steps in the computer booting process?	POST, BIOS/UEFI initialization, OS loading, driver loading	3	
3	Describe three methods of data backup and their advantages.	Local backup, cloud backup, network backup - availability and security	3	
4	How do you ensure data integrity when transferring files between different media?	Verify checksums, use error-checking, confirm file sizes match	3	
5	Explain the difference between straight-through and crossover network cables.	Straight for device-to-switch, crossover for device-to-device connections	3	
6	What are the essential elements of network troubleshooting methodology?	Identify problem, establish theory, test solutions, document results	3	
7	Name three online collaboration tools and their primary functions.	Video conferencing, cloud storage, project management platforms	3	
8	What safety measures should be observed when performing computer maintenance?	PPE usage, power disconnection, proper tool handling, ventilation	2	
9	How do you optimize computer performance through software management?	Remove unnecessary programs, update software, disk cleanup, defragmentation	2	
	Total Oral		25	

8. SUMMARY OF ASSESSMENT

S/N	SECTION	Total Marks	Marks Awarded
1.	Computer Essentials & Operations	70	
2.	Database, Presentation & Networking	60	
3.	Maintenance & Troubleshooting	55	
4.	Oral Assessment	25	
5.	Grand Total	210	

COMPETENCY DETERMINATION

The candidate was found to be:

☐ Competent

☐ Not yet Competent

(Please tick as appropriate)

The candidate is competent if the candidate obtains at least 50% (105 marks out of 210)

Assessor Signature: _____ **Date:** _____

Candidate Signature: _____ **Date:** _____

Additional Notes:

- All practical work must be completed within the allocated time
- Candidates must demonstrate proper workplace safety throughout all sessions
- Digital portfolios of work should be maintained for verification
- Any queries during assessment should be clarified with the assessor immediately